# EBS Volume Management And Backup Strategies

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**Partition:**

a partition is a logical division of a physical storage device

1. partition makes backing up easier
2. separating the user data from the system data
3. also used for organising the data
4. To improve the security and stability
5. To use different filesystems (like ext4, NTFS, etc.)

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**Common Linux partitions**

- **/** (root)
  Main partition where the OS is installed
- **/home**
  Stores user files and personal data
- **swap**
  Used as virtual memory when RAM is full
- **/boot**
  Contains bootloader and kernel files

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**Example**

A 500GB disk could be divided like:

- 100GB → / (system)
- 350GB → /home (user data)
- 50GB → swap

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**Types of partitions**

- Primary partition – main partition (bootable)
- Extended partition – container for logical partitions
- Logical partition – subdivisions inside extended

**Note:**

1. we can create up to 4 partitions
2. out of 4 partitions, 3 are primary partitions and 1 is extended partition also we can create all 4 partitions as a primary partition but only in MBR

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**How Linux identifies partitions**

Linux names them like:

- /dev/sda → whole disk
- /dev/sda1, /dev/sda2 → partitions

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**Practical: Create a Partion of the volume**

**1.create a instance**

1. Name ----------> webserver
2. Ami ----------> amazon linux
3. Instance type ---> t3. micro
4. Key name --------> ubuntu
5. Create a instance

**2.Create A Volume Of 10GB**

1. Create a volume – 10GB
2. Availability zone – 1a
3. Create a volume

**3.Attach volume to a instance**

1. Click on the volume
2. Attach a volume
    1. Availability zone – 1a
    2. Instance – webserver
    3. Device name - /dev/sdb
    4. Attach volume

**4.Create a partition of the volume & mount the volume**

1. Access the instance
2. Create a partition
3. lsblk

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```
[root@ip-172-31-41-251 ~]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
[root@ip-172-31-41-251 ~]#
```

**Note:**

1. nvme0n1 (8G) -------> system volume / disk
2. nvme0n1p1 ----------> system partition
3. nvme1n1 (10G) -----> Extra attached volume

4. **create a partition**
   fdisk /dev/nvme1n1

```
[root@ip-172-31-41-251 ~]# fdisk /dev/nvme1n1

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0xd145a473.

Command (m for help):
```

5. press m ---> (for help)
6. press n ----> (for create a partition)
7. press p ----> (for create a primary partition)
8. press 1 ----> (for creating a first partition)
9. first sector -> Empty
10. last sector --> +4G
11. press w -----> (for saving the partition table)

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1):
First sector (2048-20971519, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519): +4G

Created a new partition 1 of type 'Linux' and of size 4 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@ip-172-31-41-251 ~]#
```

12.lsblk

```
[root@ip-172-31-41-251 ~]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
└─nvme1n1p1   259:5    0   4G  0 part
[root@ip-172-31-41-251 ~]#
```

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**13.Create 2nd partition**

1. fdisk /dev/nvme1n1
2. press m ---> (for help)
3. press n ----> (for create a partition)
4. press p ----> (for create a primary partition)
5. press 2 ----> (for creating a 2nd partition)
6. first sector -> Empty
7. last sector --> +4G
8. press w -----> (for saving the partition table)

```
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
└─nvme1n1p1   259:5    0   4G  0 part
[root@ip-172-31-41-251 ~]# fdisk /dev/nvme1n1

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2):
First sector (8390656-20971519, default 8390656):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8390656-20971519, default 20971519): +4G

Created a new partition 2 of type 'Linux' and of size 4 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@ip-172-31-41-251 ~]#
```

14.lsblk

```
[root@ip-172-31-41-251 ~]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
├─nvme1n1p1   259:8    0   4G  0 part
└─nvme1n1p2   259:9    0   4G  0 part
[root@ip-172-31-41-251 ~]#
```

**Note:**

1. nvme1n1 ----------> attach disk (10GB)
2. nvme1n1p1 -------> partition1 (4GB)
3. nvme1n1p2 -------> partition2 (4GB)

## 15.Assign a filesystem to partition

mkfs.ext4 /dev/ nvme1n1p1

```
[root@ip-172-31-41-251 ~]# mkfs.ext4 /dev/nvme1n1p1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1048576 4k blocks and 262144 inodes
Filesystem UUID: 97d19d5c-fc64-44bd-81c8-c39f97a7a1aa
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@ip-172-31-41-251 ~]#
```

**16.To check the filesystem, assign or not**

1.blkid

```
[root@ip-172-31-41-251 ~]# blkid
/dev/nvme0n1p1: LABEL="/" UUID="1bc1d419-553e-4a88-91ca-fa9f67d175e5" BLOCK_SIZE="4096" TYPE="xfs" PARTLABEL="Linux" PAR
TUUID="d947e621-1922-4589-ae28-4f8b5d16efa3"
/dev/nvme0n1p128: SEC_TYPE="msdos" UUID="BAA9-6338" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI System Partition" PARTUU
ID="bfedbd6a-2a20-4781-8200-7c5f2d0d5a0b"
/dev/nvme0n1p127: PARTLABEL="BIOS Boot Partition" PARTUUID="f22ac85f-cadc-4b72-ad8a-bd647bb2f0b8"
/dev/nvme1n1p2: PARTUUID="d145a473-02"
/dev/nvme1n1p1: UUID="97d19d5c-fc64-44bd-81c8-c39f97a7a1aa" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="d145a473-01"
[root@ip-172-31-41-251 ~]#
```

**17.Mount the partition (Temporary Mounting)**

1.mount /dev/nvme1n1p1 /mnt

**18.lsblk**

```
[root@ip-172-31-41-251 ~]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
├─nvme1n1p1   259:8    0   4G  0 part /mnt
└─nvme1n1p2   259:9    0   4G  0 part
[root@ip-172-31-41-251 ~]#
```

**19.To check disk space usage along with filesystem types**

1.df -Th

```
[root@ip-172-31-41-251 ~]# df -Th
Filesystem       Type      Size  Used Avail Use% Mounted on
devtmpfs         devtmpfs  4.0M     0  4.0M   0% /dev
tmpfs            tmpfs     459M     0  459M   0% /dev/shm
tmpfs            tmpfs     184M  460K  183M   1% /run
/dev/nvme0n1p1   xfs       8.0G  1.6G  6.4G  20% /
tmpfs            tmpfs     459M     0  459M   0% /tmp
/dev/nvme0n1p128 vfat       10M  1.3M  8.7M  13% /boot/efi
tmpfs            tmpfs      92M     0   92M   0% /run/user/1000
/dev/nvme1n1p1   ext4      3.9G   24K  3.7G   1% /mnt
```

**Note:**

1. At this time, we mount the partition temporary to mnt
2. After reboot, it will disappear.

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## PERMANENT MOUNTING

1.blkid

{Copy the UUID of nvme1n1p1 from your partition}

```
[root@ip-172-31-41-251 ~]# blkid
/dev/nvme0n1p1: LABEL="/" UUID="1bc1d419-553e-4a88-91ca-fa9f67d175e5" BLOCK_SIZE="4096" TYPE="xfs" PARTLABEL="Linux" PAR
TUUID="d947e621-1922-4589-ae28-4f8b5d16efa3"
/dev/nvme0n1p128: SEC_TYPE="msdos" UUID="BAA9-6338" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI System Partition" PARTUU
ID="bfedbd6a-2a20-4781-8200-7c5f2d0d5a0b"
/dev/nvme0n1p127: PARTLABEL="BIOS Boot Partition" PARTUUID="f22ac85f-cadc-4b72-ad8a-bd647bb2f0b8"
/dev/nvme1n1p2: PARTUUID="d145a473-02"
/dev/nvme1n1p1: UUID="97d19d5c-fc64-44bd-81c8-c39f97a7a1aa" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="d145a473-01"
[root@ip-172-31-41-251 ~]#
```

2.vim /etc/fstab

UUID=97d19d5c-fc64-44bd-81c8-c39f97a7a1aa  /mnt  ext4  defaults 0  2

3.cat /etc/fstab

```
[root@ip-172-31-41-251 ~]# cat /etc/fstab
#
UUID=1bc1d419-553e-4a88-91ca-fa9f67d175e5     /          xfs    defaults,noatime  1   1
UUID=BAA9-6338        /boot/efi       vfat    defaults,noatime,uid=0,gid=0,umask=0077,shortname=winnt,x-systemd.automount 0 2

UUID=97d19d5c-fc64-44bd-81c8-c39f97a7a1aa     /mnt        ext4   defaults 0   2
[root@ip-172-31-41-251 ~]#
```

4. Update The Partition

mount -a

5.lsblk

```
[root@ip-172-31-41-251 ~]# lsblk
NAME          MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1       259:0    0   8G  0 disk
├─nvme0n1p1   259:1    0   8G  0 part /
├─nvme0n1p127 259:2    0   1M  0 part
└─nvme0n1p128 259:3    0  10M  0 part /boot/efi
nvme1n1       259:4    0  10G  0 disk
├─nvme1n1p1   259:8    0   4G  0 part /mnt
└─nvme1n1p2   259:9    0   4G  0 part
```

**Note:**

- UUID=abcd-1234 → your actual UUID of Partition
- /mnt → mount point
- ext4 → filesystem
- 0 → no dump
- 2 → correct for non-root partition (priority)

**Explanation:**

**1.Why we use 0 In permanent mounting (Dump / Backup)**

- 0 → Do NOT take backup
- 1 → Take backup (rarely used today)

{Everyone uses the almost 0 because modern systems use other backup tools}

**2.Why we use 2 when permanently mounting the partition**

a) 0 ---------> Do not check
b) 1 ---------> Check First (root /)
c) 2 ---------> Check After root

After system starts

1. Root partition (/) is checked first → 1
2. Other partitions (like /mnt) are checked later → 2

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**SIMPLE STEPS TO CREATE A PARTITION**

1. create a partition (fdisk)
2. attach a filesystem (mkfs)
3. mount the partition (mount)
4. check the disk usage (df -Th)

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**COMMANDS**

1. fdisk -------> Used to create, delete & manage the disk partitions on system
2. mkfs -------> Used to assign a filesystem to a new partition
3. blkid -------> Shows UUID and filesystem type of partitions
4. lsblk -------> Shows all disks and partitions in tree format
5. mount ----> Used to attach a partition to a directory
6. df -Th -----> Shows disk usage along with filesystem type

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## DIFFERENCE BETWEEN THE EXT4 & XFS

**EXT4 FILESYSTEM**

ext4 = Old, stable, widely used filesystem

- Best for small to medium files
- Very stable & reliable
- Supports journaling (prevents data loss)
- Easy to recover & repair
- Slower with very large data
- Example: OS disks, small applications, general usage

**XFS FILESYSTEM**

xfs = High-performance filesystem

- Best for large files & big data
- Very fast performance
- Supports parallel I/O (multi-threading)
- Scales well for large storage
- Hard to shrink filesystem
- Recovery is more complex than ext4
- Example: Databases, logs, big storage servers

**Note:**

1. ext4 = Safe + Stable
2. xfs = Fast + Powerful

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## DIFFERENCE BETWEEN THE MBR AND GPT

**MBR (Master Boot Record)**

**MBR = Old partition system**

- Supports only up to 4 partitions
- Maximum disk size: 2 TB
- Stored in one location (risk if corrupted)
- Works with older systems (BIOS)

**GPT (GUID Partition Table)**

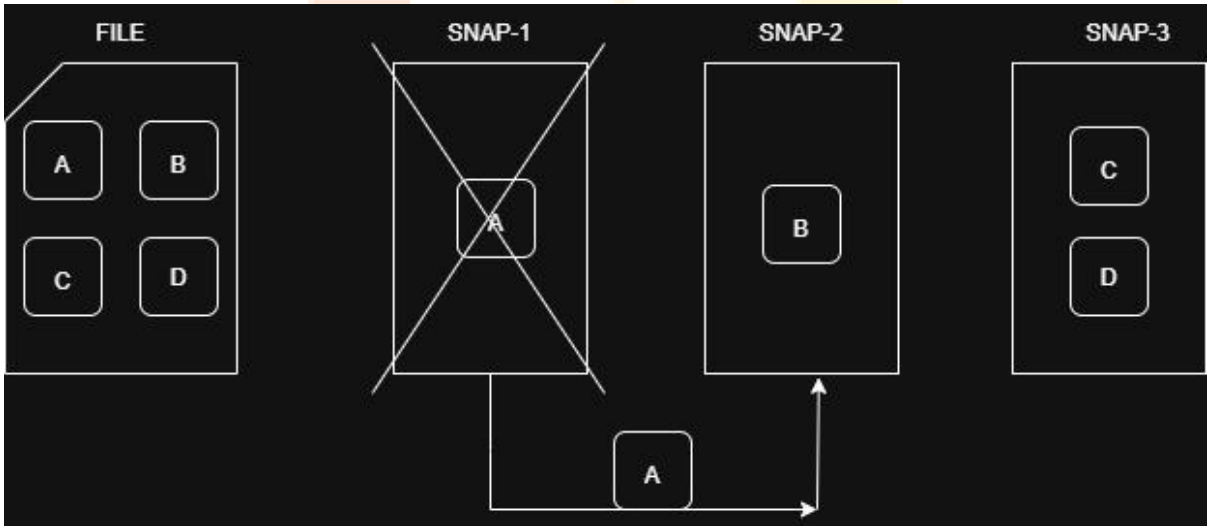
**GPT = Modern partition system**

- Supports 128 partitions (or more)
- Supports very large disks (more than 2 TB)
- Has backup (more safe)
- Works with modern systems (UEFI)

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**SNAPSHOT**

1. Snapshot is a point in time copies or point in time images of our EBS volume
2. With the help of snapshot, we can take easily backup of our EBS volume
3. EBS volumes are AZ specific While the Snapshots are region specific
4. Using snapshots, we can create a new EBS volume in any Availability Zone (AZ) within the same region.
5. As well as we can also share the snapshot from one account to another account
6. In AWS, we can create up to 10,000 EBS snapshots per region. (By Default)
7. Snapshots are incremental data backups
8. Any data written in the EBS volume after the snapshots should not be included. It will be included after the incremental updates (means after taking another snapshot)
9. EBS snapshots are stored on Amazon S3, however you cannot access them directly you can only access them through EC2 APIs.
10. you can create a snapshot to an EBS volume of the same or longer size than the original volume size, from which the snapshot was initially created. But we can't create the EBS volume less than the actual size of the snapshots.




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**TAKE BACKUP OF EBS VOLUME (Manually)**

**1.Create A Instance**

1. name ----------> webserver
2. ami ----------> ubuntu
3. key name ---------> ubuntu.pem
4. create a instance

**2.Create A Snapshot**

1. go to volume -> click on volume -> create a snapshot
2. resource type -> volume
3. description -> my root volume snapshot
4. create a snapshot

**3.Click On Snapshot**

1.click on snapshot

{You can create a image as well as volume from these snapshot}

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**TAKE BACKUP OF EBS VOLUME (automatically)**

**1.Go To Lifecycle Manager**

1. create lifecycle policy -> custom policy
2. Target resource types -> instance
3. Target resource tags
   a. key -> Name
   b. value -> ubuntu
4. 3.policy status -> enabled
5. Frequency -> Daily
6. 5.Every -> Hour
7. Starting at -> 9:00
8. Retention type -> 10
9. Review policy

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**HOW TO COPY SNAPSHOT FROM ONE REGION TO ANOTHER REGION**

**1.click on snapshot**

1. click on action
2. click on copy snapshot
3. Destination Region -> ap-southeast-1
4. Copy snapshot

**2.Go to Singapore region**

1. click on snapshot
2. you get the snapshot

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**HOW TO SHARE SNAPSHOT FROM ONE ACCOUNT TO ANOTHER ACCOUNT**

**1.Click On Snapshot**

1. click on actions
2. Snapshot setting -> Modify permissions
3. Sharing options -> private
4. Add account -> enter another account ID
5. Click on these account ID
6. Modify permissions

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## Tasks:

1. How to create a ami from snapshot
2. How to create a volume from snapshot in another AZ
3. How to copy snapshot from one region to another region
4. How to share snapshot from one account to another account
5. Create a ami of existing instance and create a new instance from these ami
6. Attach a new volume to instance, mount this volume and create some data in this volume, then detach this volume and create a new instance and attach this volume to new instance and mount this volume and verify whether the data persists.
7. Create a snapshot of a volume, delete the volume, create a new volume from the snapshot, attach it to an instance, and verify the data.

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